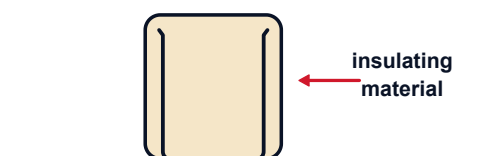


Thermal insulation

A student compares how well different materials insulate a beaker of hot water.

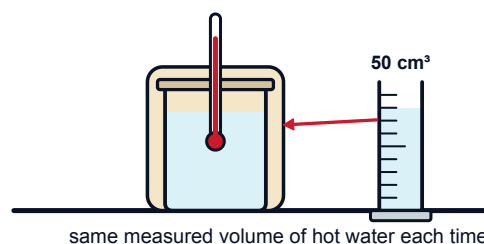
- 1** Wrap a beaker in one of the insulating materials.

wrap the beaker in an insulating material

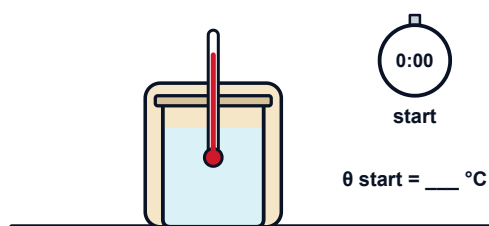


- 2** Pour in a measured volume of hot water and fit a lid with a thermometer.

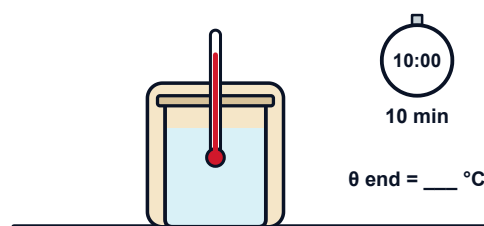
lid with thermometer



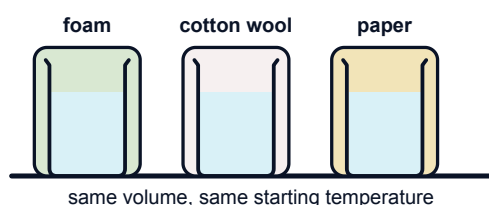
- 3** Record the starting temperature and start a stopwatch.



- 4** Record the temperature again after 10 minutes.



- 5** Repeat with each material, using the same volume and starting temperature of water.



- 6** Compare the temperature falls: the smallest fall shows the best insulator.

material	start	end	fall
foam			
cotton wool			
paper			

fall = start – end

smallest fall = best insulator